

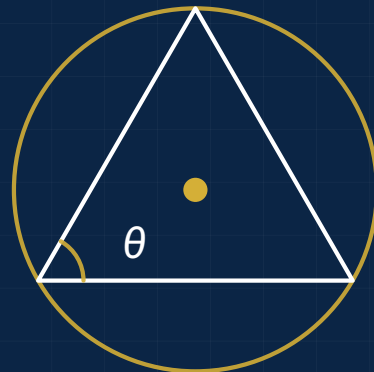


DR ESLAM AHMED
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CHAPTER 4 • GEOMETRY & MEASURES • TOPIC 11

Sine & Cosine Rules & Area

For triangles without a right angle. A side and its opposite angle → sine rule. Two sides and the angle between → cosine rule and $\frac{1}{2}ab \sin C$.



4

VARIANTS

17

PAST-PAPER PARTS

43

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Area $\frac{1}{2}ab \sin C$

V2 The sine rule

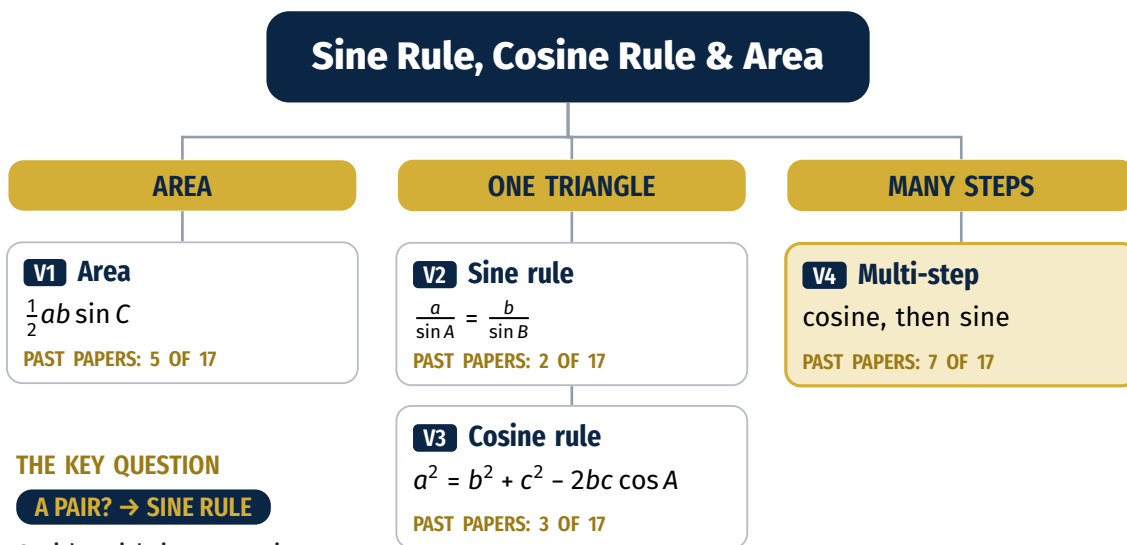
V3 The cosine rule

V4 Multi-step problems

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Sine & cosine rules at a glance



Your notes


THE KEY QUESTION

A PAIR? → SINE RULE

A side with its opposite angle known: sine. Otherwise: cosine.

★ How often does each variant come up?

16 past-paper questions (Jan 2020 → Nov 2025) hold 17 parts; 43 practice parts — grouped by type.

	PAST PAPERS (OUT OF 17)	PRACTICE PARTS
V1 Area half $ab \sin C$	5	10
V2 The sine rule	2	9
V3 The cosine rule	3	10
V4 Multi-step problems	7	14

WHAT THIS TELLS YOU On every paper (Q14–25, 3–6 marks), usually as two triangles joined: find a shared side with one rule, then use it in the other triangle. The formulas are on the formula sheet.

BEFORE YOU START

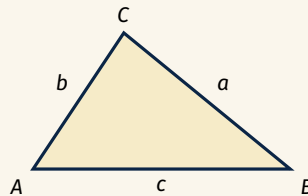
The big idea



Label with capitals and small letters.

Angle A is **opposite** side a , angle B opposite b , angle C opposite c .

A side with its **opposite angle** both known is a **pair** — that means sine rule.



The three formulas (on the formula sheet)

Area

$$\frac{1}{2}ab \sin C$$

Two sides and the angle between.

Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

A matching pair + one more.

Cosine rule

$$a^2 = b^2 + c^2 - 2bc \cos A$$

SAS → side; SSS → angle.

Choose the rule

Two sides + angle between

cosine

Find the third side.

Three sides

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

Find an angle.

A pair + one

sine

Side or angle.



CALCULATOR FIRST, THEN CHECK

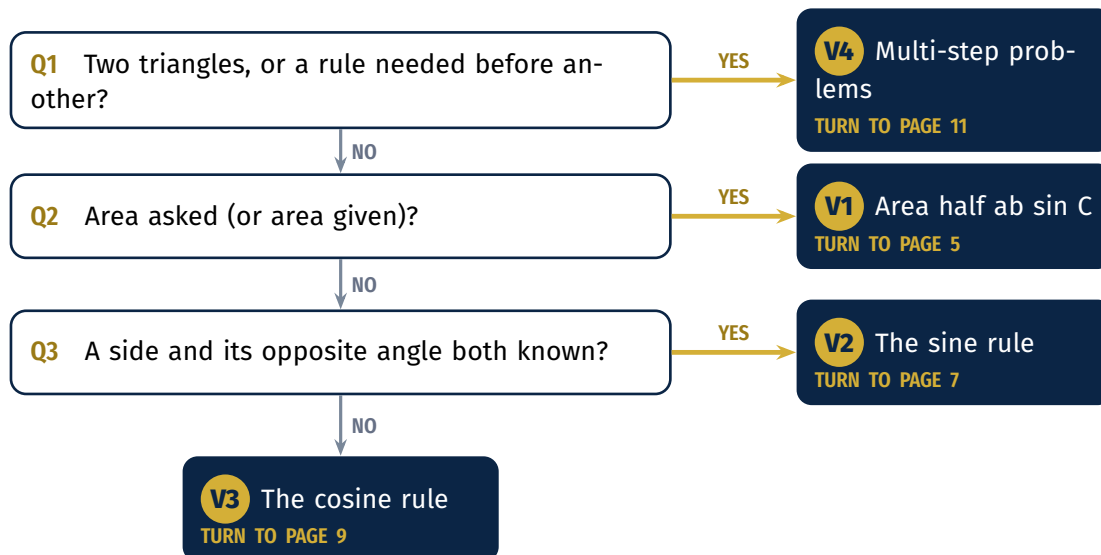
Keep the full value of a side found in step 1 (Ans key) for step 2. Obtuse angle? $\sin^{-1}$ gives the acute one: use $180 -$ it.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD



What is it?



How do I do it?



Worked example



Check yourself



When you see → you do



Watch out



Exam tip



Calculator check

V1

VARIANT 1

Area $\frac{1}{2}ab \sin C$ Two sides and the angle between them;
given the area, solve backwards.

SEEN IN PAST PAPERS

5 / 17

PRACTICE QUESTIONS: 10



Your notes

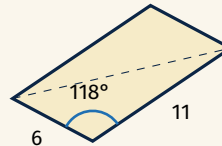
Came up in: Jan 2022 P1H Q9 • Jun 2022 P2HR Q15 • May 2023 P1HR Q25 • May 2024 P1HR Q15 • Jun 2024 P2HR Q15

? What is it?

In plain words

Area = $\frac{1}{2} \times \text{side} \times \text{side}$
 $\times \sin(\text{angle between them})$. No
 height needed.

Kite or parallelogram:
two equal triangles. Area given: put
 it in and solve for the side or angle.



IT LOOKS LIKE

kite ABCD

area 74

isosceles

How do I do it?

1 Angle between

The angle where the two sides meet.

2 Formula

 $\frac{1}{2}ab \sin C$.

3 Shapes

Kite / parallelogram: $\times 2$.

4 Backwards

Area given: solve for the side or $\sin C$.
ANGLE BETWEEN THE SIDES Not any angle.

V1 Area $\frac{1}{2}ab \sin C$

VARIANT 1 · SEE IT DONE, THEN TRY


 Your notes
 WHEN YOU SEE → YOU DO

“Calculate the area of the kite” → **Two congruent triangles:** $2 \times \frac{1}{2}ab \sin C$

“The area of triangle EFG is 74 cm^2 .
Work out EF” → $\frac{1}{2}x^2 \sin 130^\circ = 74$

An isosceles triangle with three sides → **Height by Pythagoras, or find an angle first**

 WORKED EXAMPLE **PAST PAPER**

June 2022 · Paper 2HR · Q15 · 3 marks

Kite ABCD: $AB = 6 \text{ cm}$, $BC = 11 \text{ cm}$, angle $ABC = 118^\circ$. Calculate the area of the kite (3 s.f.).

STEP 1 kite = 2 triangles ABC, ADC ◀ symmetric

STEP 2 $\frac{1}{2} \times 6 \times 11 \times \sin 118^\circ$ ◀ one triangle

STEP 3 $2 \times 29.14 = 58.27$ ◀ both

ANSWER 58.3 cm^2 ✓ each triangle under $\frac{1}{2} \times 6 \times 11 = 33$ ✓ **WATCH OUT**

✗ using an angle not between the sides

✓ use angle ABC between 6 and 11

The formula needs the included angle.

 EXAM TIP

Check: area can't exceed $\frac{1}{2}ab$ (when the angle is 90°).

 Check yourself

calculator first · answers on page 13

1.1 Jan 2022 P1H

Isosceles triangle:
 $AB = AC = 17.5 \text{ cm}$,
 $BC = 28 \text{ cm}$. Area?

Answer:

1.2 May 2024 P1HR

$EF = GF$, angle
 $EFG = 130^\circ$, area 74 cm^2 .
Find EF (3 s.f.).

Answer:

1.3 Dr Eslam

Sides 7 cm and 9 cm
with 50° between them.
Area?

Answer:

V2

VARIANT 2

The Sine Rule

Use a known pair (side + opposite angle) to find another side or angle.

SEEN IN PAST PAPERS

2 / 17

PRACTICE QUESTIONS: 9



Your notes

Came up in: Jun 2022 P1H Q18 • Jun 2025 P1H Q20

? What is it?

In plain words

$\frac{a}{\sin A} = \frac{b}{\sin B}$: each side over the sine of the angle **opposite** it.

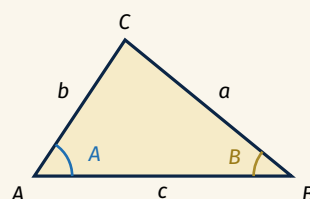
Finding a **side**: sides on top. Finding an **angle**: flip it, $\frac{\sin A}{a} = \frac{\sin B}{b}$.

IT LOOKS LIKE

$A = 40^\circ$

$B = 75^\circ$

$BC = 12$



How do I do it?

1 Pair

A side with its opposite angle, both known.

2 Second

The side or angle you want, opposite the other.

3 Write $\frac{a}{\sin A} = \frac{b}{\sin B}$ (or flipped for angles).**4 Solve**Rearrange; $\sin^{-1}$ for an angle (obtuse? 180-).**OPPOSITE PAIRS**

Each side with the angle across from it.

V2 The Sine Rule

VARIANT 2 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DOTwo angles and a side → **Sine rule for the other sides**"Work out the value of x " (an angle) → **Flipped sine rule, $\sin^{-1}$** "angle is obtuse" → $180 - \sin^{-1}(\dots)$ **WORKED EXAMPLE DR ESAM**

Sine rule • 3 marks

In triangle ABC , $A = 40^\circ$, $B = 75^\circ$ and $BC = 12$ cm. Find AC (3 s.f.).**STEP 1** $BC = a = 12$, $A = 40^\circ$ ◀ the pair**STEP 2** $AC = b$, $B = 75^\circ$ ◀ the one we want**STEP 3** $\frac{b}{\sin 75^\circ} = \frac{12}{\sin 40^\circ}$ ◀ sine rule**STEP 4** $b = \frac{12 \sin 75^\circ}{\sin 40^\circ} = 18.03$ ◀ solve**ANSWER** 18.0 cm

✓ bigger angle, bigger side ✓

**WATCH OUT**

$$\times \frac{12}{\sin 75^\circ} = \frac{b}{\sin 40^\circ}$$

$$\checkmark \frac{12}{\sin 40^\circ} = \frac{b}{\sin 75^\circ}$$

12 is opposite 40° .**EXAM TIP**

Draw an arrow from each side across to its opposite angle.

**Check yourself**

calculator first • answers on page 13

2.1

Jun 2022 P1H

$AB = 9.7$ m, $BC = 12.3$ m,
angle $B = 115^\circ$. Find
angle $C = x^\circ$ (cosine rule
first, 3 s.f.).

Answer:

2.2

Jun 2025 P1H

$AB = 9.4$, $AD = 12.8$,
angle $A = 72^\circ$; in BCD
angle $C = 54^\circ$, angle
 $BDC = 39^\circ$. Find BC .

Answer:

2.3

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$a = 10$ cm, $b = 7$ cm,
 $A = 80^\circ$. Find angle B .

Answer:

V3

VARIANT 3

The Cosine Rule

Two sides and the angle between → third side.
Three sides → any angle.

SEEN IN PAST PAPERS

3 / 17

PRACTICE QUESTIONS: 10


Your notes

Came up in: Jan 2020 P1H Q14 • Nov 2025 P1H Q19

? What is it?

In plain words

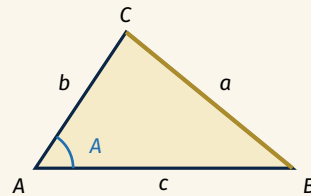
No matching pair? Use the cosine rule. It is Pythagoras with a

correction: $a^2 = b^2 + c^2 - 2bc \cos A$.

For an angle, rearrange:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

a is opposite the angle.



IT LOOKS LIKE

8

9

10 cm; parallelogram diagonal

How do I do it?

1 Which case?

SAS (side wanted) or SSS (angle wanted).

2 Label

a is opposite the angle A.

3 Substitute

Into the right form.

4 Finish

 $\sqrt{\quad}$ for a side, $\cos^{-1}$ for an angle.

a OPPOSITE A Then square root or $\cos^{-1}$.

V3 The Cosine Rule

VARIANT 3 · SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

Three sides given, "work out angle BAC"

$$\rightarrow \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

"length of the diagonal" with two sides and an angle

$$\rightarrow a^2 = b^2 + c^2 - 2bc \cos A$$

Obtuse angle given

→ **cos is negative: the term adds****WORKED EXAMPLE PAST PAPER**

November 2025 · Paper 1H · Q19 · 3 marks

AB = 8 cm, BC = 10 cm, CA = 9 cm. Work out angle BAC (1 d.p.).

STEP 1 $a = BC = 10$ (opposite A) ◀ label**STEP 2** $\cos A = \frac{8^2 + 9^2 - 10^2}{2 \times 8 \times 9}$ ◀ rearranged cosine rule**STEP 3** $= \frac{45}{144}$ ◀ simplify**STEP 4** $A = \cos^{-1}(0.3125) = 71.79 \dots$ ◀ inverse**ANSWER** 71.8°

✓ largest side opposite the largest angle? 10 is opposite A ✓

**WATCH OUT**

$$\times \cos A = \frac{10^2 + 9^2 - 8^2}{\dots}$$

$$\checkmark \frac{8^2 + 9^2 - 10^2}{\dots}$$

Subtract the side opposite the angle.

**EXAM TIP**

Write the cosine rule in letters first, then substitute.

**Check yourself**

calculator first · answers on page 13

3.1

Jan 2020 P1H

Parallelogram: $EF = 9.3$,
 $FG = 14.7$, angle
 $EFG = 106^\circ$. (a) Area. (b)
 Diagonal EG (3 s.f.).

Answer:

3.2

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Sides 5 cm and 8 cm
 with 60° between. Third
 side?

Answer:

3.3

Dr Eslam

Sides 5, 7 and 9 cm.
 Largest angle?

Answer:

V4**VARIANT 4** MOST COMMON IN THE EXAM

Multi-Step Problems

Find a shared side in one triangle, then use it in the next; area pieces add.

SEEN IN PAST PAPERS

7 / 17

PRACTICE QUESTIONS: 14



Your notes

Came up in: Jan 2020 P1HR Q21 • Nov 2021 P2H Q18 • Jan 2022 P2HR Q16 • Jan 2022 P2H Q23 • Jun 2022 P1HR Q17 • Jun 2023 P2H

Q22 • May 2024 P1H Q19

? What is it?

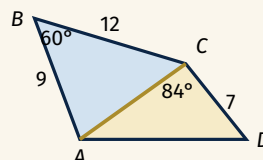
In plain words

A quadrilateral split by a **diagonal**, or a 3D or real-life shape. Start in the triangle with **enough information**; the shared side links it to the next. Keep the **full value** of each side for the next step.

IT LOOKS LIKE

quadrilateral $ABCD$

area 405 given



How do I do it?

1 Start

The triangle with 3 facts.

2 Shared side

Cosine or sine rule; keep the full value.

3 Next triangle

Use the shared side.

4 Combine

Add areas, or give the angle/side asked.

SHARED SIDE, FULL VALUE

The most common variant here.

V4 Multi-Step Problems

VARIANT 4 · SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

"Calculate the area of quadrilateral ABCD"

→ Diagonal splits it into two triangles

"Given that the area of triangle BCD is 405"

→ $\frac{1}{2}ab \sin C \rightarrow$ angle → cosine rule

"Show your working clearly"

→ Each side found on its own line



WORKED EXAMPLE PAST PAPER

November 2021 · Paper 2H · Q18 · 5 marks

Quadrilateral ABCD: $AB = 9$, $BC = 12$, angle $ABC = 60^\circ$; $CD = 7$, angle $ACD = 84^\circ$. Calculate the area of ABCD (3 s.f.).

STEP 1 $AC^2 = 9^2 + 12^2 - 2(9)(12) \cos 60^\circ = 117$ ◀ cosine rule: shared side

STEP 2 $\frac{1}{2}(9)(12) \sin 60^\circ = 46.77$ ◀ triangle ABC

STEP 3 $\frac{1}{2}\sqrt{117}(7) \sin 84^\circ = 37.65$ ◀ triangle ACD

STEP 4 $46.77 + 37.65 = 84.42$ ◀ add

ANSWER 84.4 cm^2

✓ $AC = 10.82$ kept in full ✓



WATCH OUT

× rounding AC to 10.8 first

✓ keep $\sqrt{117}$

Early rounding can move the 3 s.f. answer.



EXAM TIP

Label each new length on the diagram as soon as you find it.



Check yourself

calculator first · answers on page 13

4.1

Jan 2022 P2HR

$BC = 36$, $CD = 45$, area $BCD = 405$ (C acute);
 $AB = 19$, $AD = 28$. Angle ABD (1 d.p.)?

Answer:

4.2

May 2024 P1H

$BC = 12.8$, angle $C = 62^\circ$,
angle $BDC = 40^\circ$;
 $AB = 15.2$, $AD = 13.4$.
Angle $A = x^\circ$?

Answer:

4.3

Jan 2022 P2H

$PQ = 1.6$, $PR = 4.2$, angle $R = 18^\circ$, angle Q obtuse.
Area of PQR (3 s.f.)?

Answer:

CHECK YOURSELF

Answers



Your notes

Every answer below was checked in Python (check_answers.py).

1.1 147 cm^2

1.2 13.9 cm

1.3 24.1 cm^2

2.1 $x = 28.2$

2.2 10.4 cm

2.3 43.6°

3.1 131 cm^2 ; 19.4 cm

3.2 7 cm

3.3 95.7°

4.1 83.9°

4.2 $x = 75.6$

4.3 1.98 cm^2

THE WHOLE TOPIC ON ONE PAGE

Sine & cosine cheat sheet



Your notes
V1 Area 5/17

$$\frac{1}{2}ab \sin C$$

Angle between the sides.

V2 Sine rule 2/17

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

A pair + one more.

V3 Cosine rule 3/17

$$a^2 = b^2 + c^2 - 2bc \cos A$$

SAS or SSS.

V4 Multi-step 7/17

shared side

Full value; add areas.

**TOP 3 LOST MARKS**

1. side not opposite its angle ×
2. acute answer for an obtuse angle ×
3. rounding a shared side ×

BOTTOM LINE Pair → sine · no pair → cosine · $\frac{1}{2}ab \sin C$ · keep full values.